Yaqeen at IslamicEval 2026: From Contextual Detection to Retrieval-Based Verification and Canonical Correction of LLM-Generated Arabic Islamic Citations

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Abstract

Large language models are increasingly used to answer questions about religious texts, yet their fluent responses may contain misquoted, misattributed, or fabricated citations. Detecting such hallucinations requires identifying citation components, verifying their correctness, and recovering the canonical text when a quotation is incorrect. We present Yaqeen's submission to the IslamicEval 2026 shared task, which evaluates these capabilities through three subtasks: detection of Qur'anic verses, Hadith text, chains of narration, and claimed sources; identification of incorrect citation fragments; and correction of erroneous Qur'anic and Hadith quotations. We develop task-specific systems combining AraBERT-based character-level sequence labeling with retrieval-based matching, classification, and reranking. Our systems achieve 96.78% character-level Macro-F1 on Task 1, 91.07% accuracy on Task 2, and 78.02% accuracy on Task 3. The results show that combining contextual Arabic representations with retrieval-based evidence is effective for fine-grained hallucination detection and correction in Arabic Islamic content.

1 Introduction

Large language models (LLMs) are increasingly capable of answering Arabic questions, including those concerning the Qur'an and Hadith (Abdelaal et al., 2026; Brown et al., 2020; Lewis et al., 2020). However, fluent generation does not ensure textual or factual faithfulness (Ji et al., 2023). In religious content, models may alter quotations, combine texts, misattribute passages, or produce nonexistent citations, where textual authenticity is particularly important. Such errors are difficult to detect because hallucinated quotations may remain coherent while differing from the original by only a few characters or words. Moreover, citations may contain independently correct or incorrect components, such as quoted text, narration chains, and claimed sources, making response-level evaluation insufficient. IslamicEval 2026 (Alharbi et al., 2026) addresses this through three tasks: character-level citation span detection, correctness verification, and correction of incorrect Qur'anic and Hadith quotations.

We present Yaqeen, a system covering all three tasks, with three main contributions. First, we develop an offset-preserving AraBERT detector that aggregates logits across overlapping windows and reconstructs character-level citation spans. Second, we introduce evidence-based verification combining canonical similarity, token coverage, sequence agreement, claimed-source rank, and narrator compatibility. Third, we formulate correction as type-specific selective retrieval with calibrated abstention and Qur'an-specific reranking. Yaqeen achieves official test scores of 96.78% Macro F1 for span detection, 91.07% Macro Accuracy for hallucination identification, and 78.02% Macro Accuracy for correction.

2 Task Overview

IslamicEval 2026 decomposes Arabic Islamic citation verification into three tasks: (1) citation span detection, (2) hallucination identification, and (3) hallucination correction. The details of the tasks are discussed below.

Task 1: Given an LLM-generated response, the system assigns each character to one of five classes: `Ayah`, `Hadith matn`, `Isnad`, `claimed source`, or `neither`. Performance is measured by character-level Macro-F1. Task 2: For each citation fragment, the system determines whether it is correct or incorrect. An *Isnad* or claimed source is evaluated only when its corresponding matn is correct; otherwise, it is excluded. Performance is measured by accuracy. Task 3: Given an incorrect Ayah or Hadith matn, the system recovers its corre-

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sponding text from the shared resources, including prescribed writing and diacritics, or returns 'mistake' when no valid correction can be established. Performance is measured by accuracy.

2.1 Dataset

The IslamicEval 2026 dataset contains 4,706 Arabic training responses with 15,826 citation annotations and 32,779 labeled segments, plus 484 validation responses. The shared-task organizers also provided the retrieval resources, comprising 6,236 Qur'anic verses and 24,793 Arabic Hadith matn–isnad records from six major Hadith books.

3 Related Work

Arabic pretrained language models have shown strong performance across Arabic NLP tasks, including sequence labeling and text classification (Antoun et al., 2020; Melhem et al., 2024; Nasr and Ben HajHmida, 2024), motivating our use of AraBERT for span detection. Retrieval-based methods have also been successfully applied to Arabic and Qur'anic question answering, where relevant passages are identified from trusted collections before extracting an answer (Alnefaie et al., 2023). Such grounding is particularly important for religious texts, where minor textual or attribution errors can affect correctness. Following this direction, we ground verification and correction in authoritative Qur'anic and Hadith collections, combining character n-gram retrieval with evidence-based classification. Rather than generating corrections freely, our approach selects them directly from the reference collections to preserve their original wording.

4 Methodology

Our submission consists of three task-specific systems. We use pretrained Arabic language models where contextual understanding is important, and retrieval-based methods for verifying cited texts. The systems are designed independently, but Tasks 2 and 3 share the same general retrieval and matching components.

4.1 Task 1: Span Detection

We use `aubmindlab/bert-base-arabertv02` (Antoun et al., 2020) for five-class sequence labeling. Responses are tokenized into overlapping windows, and each token is assigned the majority character-level label within its offset interval; tokens without valid offsets are ignored. We freeze the embedding and lower Transformer layers and fine-tune the top four layers and classification head for two epochs using AdamW (Loshchilov and Hutter, 2019), with a learning rate of 3×10^{-5} and batch size 8. At inference, logits from overlapping windows are averaged at the character level before selecting the final label. Whitespace omitted during tokenization inherits a non-background label only when both adjacent characters agree. Consecutive characters with the same label are then reconstructed into non-overlapping spans using exclusive end offsets. Finally, we apply validation-selected minimum lengths of four characters for general spans and ten for Ayah and Isnad spans to suppress short spurious detections.

4.2 Task 2: Hallucination Identification

We perform evidence-based verification by retrieving candidate texts and combining textual and citation-level evidence to classify each fragment as correct or incorrect. Inputs are normalized by removing diacritics, Qur'anic recitation marks, tatweel, and punctuation; standardizing alif and hamza variants; and normalizing whitespace. Qur'anic and Hadith candidates are indexed with character n-gram TF–IDF and retrieved by cosine similarity. Qur'anic fragments are matched against the complete verse collection, while Hadith retrieval uses character 3–5-grams.

For the top candidate, we compute retrieval similarity, token coverage, and sequence agreement:

$$\text{coverage}(q, c) = \frac{|T(q) \cap T(c)|}{|T(q)|},$$

where $T(x)$ is the normalized token set. Coverage measures content support, while sequence agreement captures consistency in token order.

For Ayah and matn verification, we train separate histogram gradient-boosting classifiers. Features include retrieval similarity, coverage, sequence agreement, length, and source availability, with Ayah additionally using claimed-verse evidence and matn using narrator and chain evidence. Predictions are considered correct above thresholds of 0.65 for Ayah and 0.70 for matn.

Claimed-source verification combines sparse logistic regression with retrieved citation evidence. Qur'anic chapter names and verse numbers are parsed and checked against retrieval, while Hadith sources are compared with sources associ-

ated with similar matn. Isnad verification similarly combines sparse classification, normalized narrator overlap, and retrieved-matn evidence after removing transmission verbs, honorifics, and generic terms. Because a Hadith may have multiple valid chains, missing narrator overlap alone is not treated as sufficient evidence of an incorrect Isnad.

Finally, following the task dependency structure, Isnad and claimed-source verification is performed only when the associated matn is classified as correct, preventing attribution evidence from compensating for an incorrect quotation.

4.3 Task 3: Hallucination Correction

We formulate Task 3 as retrieval and reranking rather than direct generation. Separate character n-gram TF–IDF retrievers search the Qur'an and Hadith collections, augmented with deduplicated training corrections. For each query, we retain the top candidates and use retrieval scores and their margin to estimate ambiguity.

A type-specific histogram gradient-boosting classifier decides whether to return a correction or abstain, using retrieval scores, score margin, length features, and source-correctness indicators. Acceptance thresholds are 0.50 for Ayah and 0.70 for matn, reflecting the greater variation and incomplete coverage of Hadith texts.

For accepted Ayah predictions, we rerank the top 20 candidates using retrieval rank and score, character and token similarity, word overlap, and length-based features. For matn, validation ablation showed that reranking reduced performance, so the retrieval-and-abstention decision is used directly. Reranking is also skipped when a Qur'anic source has already been resolved from a correct structured citation.

Finally, corrections are copied directly from the retrieved collection rather than generated, preserving the required wording and diacritics.

5 Results

5.1 Baseline Comparison

Table 1 compares our final systems with simple development-set baselines. Task 1 shows the largest gain, with AraBERT-based span detection improving Macro F1 by 21.64 points over lexicon and structural rules. Task 2 improves Macro Accuracy by 7.14 points through retrieval evidence and type-specific classifiers, while Task 3 gains 5.30 points using learned abstention and Ayah reranking. Overall, learned task-specific mechanisms consistently outperform fixed rules and thresholds.

Table 1: Development-set comparison between baseline methods and Yaqeen.

Task	Baseline Method	Metric	Baseline	Yaqeen
Task 1	Lexicon and structural rules	Macro F1	74.18	**95.82**
Task 2	Canonical similarity thresholds	Macro Acc.	85.92	**93.06**
Task 3	Retrieval with fixed thresholds	Macro Acc.	72.64	**77.94**

Table 2: Development and official test-set performance.

Task	Component	Metric	Dev	Test
Task 1	Overall	Macro F1	95.82	96.78
Task 2	Overall	Macro Acc.	93.06	91.07
	Ayah	Acc.	94.70	92.00
	Matn	Acc.	93.37	84.43
	Isnad	Acc.	90.00	92.26
	Claimed source	Acc.	94.17	95.60
Task 3	Overall	Macro Acc.	77.97	78.02
	Ayah	Acc.	75.00	70.44
	Matn	Acc.	80.94	84.82

5.2 Official Test Results

Table 2 reports Yaqeen's development and official test-set performance, including component-level results where available. Task 1 achieves the highest overall performance, improving from 95.82% Macro F1 on the development set to 96.78% on the test set. This suggests that the span-detection model generalizes well beyond the development data. Remaining errors are mainly concentrated around difficult boundaries and short or ambiguous segments, as discussed in Section 6. For Task 2, Macro Accuracy decreases from 93.06% on development data to 91.07% on the test set. This reduction is driven primarily by matn verification, which decreases from 93.37% to 84.43%. In contrast, claimed-source and Isnad verification improve to 95.60% and 92.26%, respectively, while Ayah verification remains relatively stable at 92.00%. The lower matn performance reflects greater textual and transmission variation in Hadith, where surface similarity alone may not establish correctness. Task 3 remains stable overall, achieving 77.97% Macro Accuracy on the development set and 78.02% on the test

set. However, component-level performance differs: Ayah correction decreases from 75.00% to 70.44%, whereas matn correction improves from 80.94% to 84.82%. The higher matn correction score does not necessarily mean that Hadith is easier to correct. Abstention is more prevalent among incorrect matn training instances than Ayah instances (75.53% vs. 54.71%), which favors the higher matn acceptance threshold. Hadith evaluation is also diacritic-insensitive, whereas Qur'anic evaluation preserves more orthographic information. In addition, similar or overlapping verses can make the intended Ayah difficult to identify. The difference therefore likely reflects the data distribution and evaluation setting rather than the intrinsic difficulty of the two text types.

6 Error Analysis

We manually examined 60 validation errors, with 20 examples from each task, to identify recurring failure modes. Task 1 errors mainly involve citation boundaries, Task 2 errors reflect the gap between textual plausibility and contextual correctness, and Task 3 errors arise from ambiguity in recovering the intended source text.

6.1 Task 1: Citation-Boundary Ambiguity

Most Task 1 errors occur at the boundaries between `Isnad`, `matn`, and `claimed source`. The model often identifies the correct citation region but assigns boundary tokens to the wrong component. For example, the gold `Isnad` ends with عن ابن عمر رضي الله عنهما قال:, while the model extends it into كان رسول الله, consequently truncating the `matn`. Reporting verbs such as روى are similarly assigned inconsistently to the source or `Isnad`. Very short `Isnad` spans are also frequently missed. Expressions such as قال: and حديث: provide little contextual evidence and can occur in ordinary prose. These errors indicate difficulty reproducing precise boundaries when adjacent citation components are linguistically continuous.

6.2 Task 2: Plausibility Is Not Correctness

Task 2 errors often arise when textual plausibility is mistaken for contextual correctness. This is particularly evident for `Isnad` and `claimed source` fragments: valid-looking attributions such as رواه مسلم were sometimes accepted despite being incorrect for the associated quotation. Conversely, genuine quotations were rejected when orthographic or transmission variants reduced similarity to the retrieved text. Thus, surface similarity can cause both false acceptance and false rejection. Verification requires establishing the relationship between a citation and its specific source, rather than only determining whether the fragment resembles an authentic Islamic text.

6.3 Task 3: Ambiguous Recovery

Task 3 errors are dominated by conservative abstention and incorrect selection among plausible candidates. Abstention occurs when a corrupted quotation retains too little text for retrieval to reach its confidence threshold. For example, a fragment containing only كلوا من الطيبات shares an anchor with a Qur'anic verse, but the remaining corruption provides insufficient evidence for confident correction. When an incorrect candidate is selected, a short shared phrase can dominate the ranking. For example, الله لا إله إلا هو على كل شيء قدير led to a verse sharing على كل شيء قدير, although the intended source was different. Similarly, a quotation combining material from multiple verses was assigned to the candidate with greater lexical coverage rather than the intended verse. These errors reveal a precision–recall trade-off: stricter acceptance reduces unsupported corrections but increases abstentions. More reliable correction therefore requires ranking evidence across the entire quotation rather than relying on short overlapping phrases.

7 Conclusion

In this work, we presented Yaqeen's submission to IslamicEval 2026. Our systems achieve 96.78% character-level Macro-F1 on Task 1, 91.07% accuracy on Task 2, and 78.02% accuracy on Task 3. The results show that span detection is comparatively robust, while verification and correction are more challenging because correctness depends on the relationship between a citation and its canonical source. Our error analysis highlights citation-boundary ambiguity, Hadith variation, and short overlapping phrases as recurring sources of failure. Future improvements should therefore focus on source-aware retrieval and verification methods that use the full citation context to distinguish plausible but incorrect quotations from authentic ones.

References

Ali Abdelaal, Mohammed Nader Al Haffar, Mahmoud Fawzi, and Walid Magdy. 2026. Islamicmmlu: A benchmark for evaluating llms on islamic knowledge. *arXiv preprint arXiv:2603.23750*.

Rahaf Alharbi, Abdulelah Alturki, Watheq Mansour, Rana Malhas, Hamdy Mubarak, Kareem Darwish, Tamer Elsayed, and Walid Magdy. 2026. IslamicEval 2026: The second shared task of capturing LLMs hallucination in Islamic content. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*.

Sarah Alnefaie, Abdullah Alsaleh, Eric Atwell, Mohammad Alsalka, and Abdulrahman Altahhan. 2023. LKAU23 at qur'an QA 2023: Using transformer models for retrieving passages and finding answers to questions from the qur'an. In *Proceedings of ArabicNLP 2023*, pages 720–727. Association for Computational Linguistics.

Wissam Antoun, Fady Baly, and Hazem Hajj. 2020. Arabert: Transformer-based model for arabic language understanding. In *Proceedings of the 4th Workshop on Open-Source Arabic Corpora and Processing Tools, with a Shared Task on Offensive Language Detection*, pages 9–15.

Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901.

Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Yejin Yeung, Marco Del Tredici, Alwyn Mok, et al. 2023. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474.

Ilya Loshchilov and Frank Hutter. 2019. Decoupled weight decay regularization. In *International Conference on Learning Representations*.

Anas Melhem, Osama Hamed, and Thaer Sammar. 2024. TAO at StanceEval2024 shared task: Arabic stance detection using AraBERT. In *Proceedings of the Second Arabic Natural Language Processing Conference*, pages 842–846, Bangkok, Thailand. Association for Computational Linguistics.

Abdelmomen Nasr and Moez Ben HajHmida. 2024. SENIT at AraFinNLP2024: trust your model or combine two. In *Proceedings of the Second Arabic Natural Language Processing Conference*, pages 428–432, Bangkok, Thailand. Association for Computational Linguistics.